

BENJAMIN RIOT-BRETÊCHER

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EDUCATION

Ph.D - Atmospheric and Oceanic Sciences, McGill University 2022 -

Supervisor: [Prof. Yi Huang](#)

Relevant Coursework: Atmospheric Dynamics, Atmospheric Radiation, Numerical Methods, Waves & Stability

ACHIEVEMENTS: Comprehensive Examination (February 2024); Proposal Examination (August 2025)

B.S - Atmospheric and Oceanic Sciences, Université du Québec à Montréal 2019 - 2022

PROFESSIONAL EXPERIENCE

Teaching Assistant 2023 -

McGill University

- [Introduction to Atmospheric Sciences - ATOC181](#) (Fall 2023, Fall 2025)
- [Science of Storms - ATOC184](#) (Winter 2024, Winter 2026)

Visitor Student Oct 2024 - Jan 2025

[School of Atmospheric Sciences, Nanjing University](#)

Treasurer Aug 2023 - Aug 2024

Council of Atmospheric and Oceanic Sciences (CAOS)

Field Campaign Involvement 2022 - Present

McGill University

- [ECALOT Field Campaign](#) (Fall 2024)
- [WHAFFERS](#) (Lead student investigator; Winter 2025)
- [GHG Montreal Project](#) (Ongoing)
- [PONEX](#) (Lead student investigator; Winter 2026)

Research Assistant May 2022 - Aug 2022

McGill University

Supervisor: [Prof. Yi Huang](#)

Research Assistant June 2021 - Aug 2021

Université du Québec à Montréal

Supervisor: [Prof. Jean-Pierre Blanchet](#)

PUBLICATIONS

- L. Liu, Y. Huang, Z. Qu, **B. Riot-Bretêcher**, J. Cole, H. Barker, et al., *Validation of EarthCARE Cloud Microphysical Profiles by Spectral Radiance Closure and Broadband Flux Analysis* (in prep)
- Z. Qu, C. Nguyen, P. Borque, A. Korolev, H. Barker, K. Ranjbar, L. Liu, Z. Mariani, M. Kacimi, M. Cholette, I. Heckman, K. Bala, N. Bliankinshtein, L. Nichman, **B. Riot-Bretêcher**, Y. Huang, J. Cole, R. Peroni, S. Holden, R. Reed, R. Crawford, D. Michelson, E. Bigras, M. Wheeler, J. Gyakum, J. Milbrandt, J. Thériault, K. Walker, J.-P. Blanchet, Y. Blanchard, *EarthCARE Commissioning Calibration/Validation Campaign in Ottawa (ECALOT)* (in prep)
- B. Pery, H. Brindley, J. Murray, S. Mosselmans, S. Panditharatne, L. Warwick, R. Hogan, Y. Huang, **B. Riot-Bretêcher**, H. Oetjen, D. Schuettmeyer, T. L'Ecuyer, [Towards ground-to-space radiative closure in the mid- to far-infrared](#), EGU-sphere, 2026 (in review for Atmospheric Measurement Techniques).
- Y. Wang, G. X. He, **B. Riot-Bretêcher**, M. D. Pelletier, Y. Huang, S. K. Loeb, [Integrating meteorological data into linear mixed models for sunlight inactivation of indicator organisms in natural lake water](#), Water Research, 2026.
- R. G. Moguel, P. M. J. Douglas, K. Ranjbar, L. Nichman, J. Asomaning, Q. Dyer-Hawes, G. Villarruel, D. Romanic, L. Liu, **B. Riot-Bretêcher**, B. Porto Ghirardi, J. Stix, M. Kalacska, O. Lucanus, J. P. Arroyo Mora, S. Ars, F. Vogel, F. Fabry, J. Gyakum, Y. Huang, [Stable isotope constraints on urban CO₂ and CH₄ emission sources in Montreal derived from aircraft, UAV, and ground measurements](#), Atmospheric Environment, 2026.
- Q. Dyer-Hawes, D. Romanic, Y. Huang, J. R. Gyakum, P. M. J. Douglas, L. Liu, **B. Riot-Bretêcher**, R. G. Moguel, J. Asomaning, J. Stix, G. Villarruel, M. Kalacska, O. Lucanus, J. P. Arroyo-Mora, K. Ranjbar, L. Nichman, F. Vogel, [Analysis of urban greenhouse gas dispersion for downtown Montréal using computational fluid dynamics and field measurements](#), Atmospheric Pollution Research, 2026.

- **B. Riot-Bret cher**, Y. Huang, L. Liu, J. Gero, [Strengthening Atmospheric Greenhouse Effect at the Arctic North Slope of Alaska Site Evidenced by Long-Term Records of the Downwelling Longwave Radiance Spectrum](#), Journal of Geophysical Research: Atmospheres, 2025.

INSTRUMENTATION AND MENTORING EXPERIENCES

Atmospheric Emitted Radiance Interferometer (AERI)

2022 - Present

McGill University

- Third blackbody (3BB) calibration tests to verify radiometric calibration accuracy and stability
- Field-of-view (FOV) mapping tests
- Interferometer swap to extend the instrument’s spectral range (E-AERI)
- Deployment in Inuvik (NWT) in the Arctic for the PONEX campaign
- Mentored Maelyn Kaya on AERI procedures (3BB test, FOV mapping, deployment/installation, long-term trend analysis) and Luke Martin on data analysis

CONFERENCES AND TALKS

- [Joint FORUM-HAWC Workshop](#) - **Poster** - Montreal, QC, Canada - October 2025
- [Gordon Research Conference on Radiation and Climate](#) - **Invited Speaker** - Lewiston, ME, USA - July 2025
- [Gordon Research Seminar on Radiation and Climate](#) - **Speaker** - Lewiston, ME, USA - July 2025
- [2nd ESA-JAXA EarthCARE In-Orbit Validation Workshop](#) - **Co-author, Presented by L. Liu** - Frascati, Italy - March 2025 Validation of EarthCARE cloud retrievals using surface spectral infrared radiances from the ECALOT campaign. (Liu, L., **Riot-Bret cher, B.** et al.)
- [Nanjing University - Atmospheric Sciences Seminar](#) - **Speaker** - Nanjing, China - December 2024
- [43rd Canadian Symposium on Remote Sensing](#) - **Attendee** - Qu bec City, QC, Canada - July 2022
- [Canadian Meteorological and Oceanographic Society](#) - **Attendee** - Remote - June 2021

AWARDS AND SCHOLARSHIP

- Dan Ștefan Petrescu Fellowship (2026 - 2027) - \$ 30,000
- McGill Graduate Excellence Fellowship (2022 - 2028)
- Graduate Mobility Award - McGill University (Oct 2024 - Jan 2025) - \$ 3,800
- Canadian Meteorological and Oceanographic Society Undergraduate Scholarship (April 2021) - \$ 1,000

SOFTWARE AND PACKAGES

- [Atmospheric Emitted Radiance Interferometer \(AERI\) data pipeline](#): raw to calibrated radiance spectra processing, quality control, data visualization, GEOMS-compliant data conversion

SKILLS

Technical Skills	MATLAB, Fortran, Python, Java, R, Bash, Linux, Radiative Transfer Model, Weather Balloon Launches, AERI Maintenance, Optimal Estimation Method
Mathematical Skills	Algebra, Calculus, Probability, Statistics
Certifications	• Data Security - Compute Ontario (June 2025) • Introduction to Advanced Research Computing - Compute Ontario (June 2025) • Introduction to Version Control Using Git - Compute Ontario (June 2025) • Scaling Up HPC Workflows - Compute Ontario (June 2025) • Career Essentials in Software Development by Microsoft and LinkedIn (Certificate ID) • Programming Foundation: Beyond the Fundamentals by LinkedIn (Certificate ID)